

# Smart tourism technologies in travel planning: The role of exploration and exploitation



C. Derrick Huang<sup>a,\*</sup>, Jahyun Goo<sup>a</sup>, Kichan Nam<sup>b</sup>, Chul Woo Yoo<sup>a</sup>

<sup>a</sup> Florida Atlantic University, Boca Raton, FL, USA

<sup>b</sup> American University of Sharjah, Sharjah, United Arab Emirates

## ARTICLE INFO

### Article history:

Received 2 March 2016

Received in revised form 8 September 2016

Accepted 28 November 2016

Available online 7 January 2017

### Keywords:

Smart tourism technologies

Smart tourism

Travel planning

Travel experience

Exploration

Exploitation

## ABSTRACT

The use of smart tourism technologies such as travel-related websites, social media, and smartphones in travel planning has been pervasive and growing. This study examines the mechanism of how travelers use these technologies to enhance travel satisfaction. By adopting the framework of exploration and exploitation and identifying the antecedents that advance and prohibit such uses, we find that the attributes of smart tourism technologies promote both explorative and exploitative use, while user's security and privacy concerns have a negative effect. In addition, explorative use has a strong influence on overall travel experience satisfaction, and exploitative use mainly enhances the transaction satisfaction.

© 2017 Elsevier B.V. All rights reserved.

## 1. Introduction

Internet and other information technologies have changed consumer behavior. In 2015, the number of Internet users was reported to reach nearly 3.2 billion in the world [79], the number of mobile users close to 4.4 billion [79], and the number of smartphone users to 1.9 billion [23]. In tourism, these technologies have played a critical role not only for the competitiveness of tourism organizations but also for the experience of tourists. Today, a vast majority of travel information search and reservations and payments during the preparation phase of a trip is done over the Internet. Even during the traveling period, Internet, smartphones, and other technologies provide travelers rich, diverse, and useful information due to easy accessibility and connection. The term smart tourism, along with the technologies that are associated with it, has become a common term to describe such interconnection, synchronization, and concerted use of different technologies for travel [32].

A plethora of technology channels and providers such as websites of traditional destinations or agencies, personal blogs, review sites, and social networks have emerged as smart tourism technologies (STTs), and their effects have been felt by all

participants in travel and tourism, ranging from travel vendors to aggregation providers to information brokers to travelers themselves. Initially, STTs have enhanced the efficiency of transaction-oriented functions such as reservation and payment. Although such transaction-related use is still important, travelers increasingly turn to this diverse array of technologies and channels to obtain more accurate, rich, comprehensive, and personalized information from the planning phase of travel to real-time experience [92]. As such, various complex marketing strategies have been developed to assist travelers in making more informed and effective decisions. For instance, the consumers' evaluation scores for hotel performance provided by the sites such as tripadvisor.com and booking.com significantly influence travelers' purchasing behavior [5,26].

With such a broad and deep impact of information technologies on travel, it is no wonder that there have been considerable research interests on smart tourism. In particular, a number of studies have been focused on the use of STTs during the planning phase of travel [e.g., 9,25,77,92,93]. Although this stream of research has supported the various types of technologies used in travel and tourism and their attributes, a further examination of the underlying mechanism with which travelers use these STTs can greatly advance our understanding of the role that these technologies play in travel planning activities to achieve satisfaction. In this study, we borrow the powerful framework of exploration and exploitation, first adopted in the context of organizational learning and later extended to information

\* Corresponding author.

E-mail addresses: [dhuang@fau.edu](mailto:dhuang@fau.edu) (C. D. Huang), [jgoo@fau.edu](mailto:jgoo@fau.edu) (J. Goo), [knam@aus.edu](mailto:knam@aus.edu) (K. Nam), [yoooc@fau.edu](mailto:yoooc@fau.edu) (C.W. Yoo).

technology use, to inspect the way that STTs are used in travel planning. We intend to examine the following research questions: (1) How the attributes of STTs motivate the different uses of STTs, and (2) How exploitative use of STTs (ETU) and explorative use of STTs (ERU) influence travel experience satisfaction (TES). To address these questions, we conduct an individual-level study using the survey data collected from a travel club in South Korea, and the results of the study offer several important contributions to the current literature on tourism. As such, our study explores the travel decision process by extending the framework of explorative and exploitative use of information technology. Furthermore, by testing the relationships between the STT attributes (STTAs) and two different STT uses, this study extends the understanding of how the use of STT leads to overall TES.

This paper is organized as follows. In the next section, we first review the previous research on STTs and their role in travel planning. We then discuss the theory of exploration and exploitation and develop the research model based on this framework. The methodology of the study, as well as the result and analysis, are presented in the ensuing sections. Finally, we discuss the theoretical and practical implication of our research, and offer suggestions for future studies.

## 2. Research background and hypotheses

### 2.1. Smart tourism technologies

Travel and tourism industries have actively adopted the Internet as a new distribution channel since the mid-1990s, and such use soon became the leading application in business-to-consumer (B2C) or E-commerce [90]. It was found that by the late 2000s travelers have been fully adapted to the use of online tools and that the use of Internet for travel-related transactions has reached a level of saturation [93]. The industry also aggressively uses emerging technologies as a marketing medium [12,64]. New applications such as social media and smartphones add to the richness of such online tourism channels. For this study, we define STTs to include all forms of online tourism applications and information sources such as online travel agents, personal blogs, public websites, company websites, social media, smartphone apps, and so on. STTs have become a major research topic in tourism, and recent studies can be classified into three major themes: understanding of online information channels [60]; website features and characteristics [41]; and new, emerging channels [32,89].

Xiang et al. [92] addressed how the role of information technology and the Internet has markedly transformed travel and tourism. They find that traditional online consumers remain unchanged with their pattern of use of online tools (e.g., hotel booking), while some groups are adopting emergent information sources (e.g., social media) and usage patterns. Although using the Internet for primary products such as airline tickets, car rental, and accommodation – the so-called e-Tourism – already reached a level of maturity as the norm for the travel and tourism industry, online searches and transactions have increasingly included secondary products such as museum tickets, shopping, or dining. To understand both the breadth and depth of STTs, No and Kim [60] examine several major online information sources and find that social networking services exhibit lower levels of these attributes than other sources, implying that travel providers should try to understand how to more effectively use such channels to maintain their competitive advantages [cf. 17,91].

The investigation of website features is the second major theme. Because websites are generally considered as the main starting point to initiate information search, the content of website remains a critical part of traveler's decision-making. In one of the

first attempts to study the web characteristics of tourism, Kaplanidou and Vogt [41] use the technology acceptance model to evaluate the influence of tourism website characteristics on perceived website usefulness in trip planning. They find that accessibility, ease of navigation, and content are the most important characteristics in users' perception of websites. Another study compares website information quality for four cities – Beijing, Hong Kong, Shanghai, and Taipei – and the potential effects on the travelers' decisions [7]. More recently, Chung et al. [18] investigate the relationships among quality, usefulness, satisfaction, and continued usage intention of destination websites as well as the relationships of these factors with the intention to visit destinations. This study shows that website information actually has an effect on travelers' destination decision.

The third theme is focused on the emerging STT channel, in particular, social media, which is considered to have the richness, unbiasedness, and responsiveness for travelers. Xiang and Gretzel [91] first investigate popular social media by using the Google search engine and list 20 unique sites important in the online tourism domain. Chung and Koo [17] study the determinants of social media usage in travel information search based on the value-based adoption model. They find that the travelers' perception of the value of social media is a primary determinant of usage. In particular, enjoyment is found to be the most important factor affecting perceived value and actual usage of social media, implying that social media sites should provide more interesting and enjoyable contents to attract travelers. Other studies have focused on diverse topics and issues regarding social media. For instance, Munar and Jacobsen [58] examine motivations for sharing tourism experiences through social media, and Kim et al. [42] study seniors' decision to revisit social networking sites for tour-related purpose.

With the growing popularity, constantly improving functionality, and the resulting favorable experiences, SSTs play an increasingly important role in travel and tourism. This study focus on one particular aspect – SSTs in travel planning – and a review of this stream of research is presented in the next section.

### 2.2. Travel planning and STTs

Currently, travel planning and decision-making have been the subject of considerable research interests. Travel planning follows a series of episodes that involve a hierarchical structure of information searches [35,63]. Drawing from several previously proposed models, travel decision-making process can include the following phases: idea formation, searching for information, evaluate alternatives for final decision, and booking [6,36,38,75]. This process model is also consistent with the six-stage search process proposed by Kuhlthau [44] and the behavior model of information-seeking patterns proposed by Ellis [22]. It is worthwhile to note that travel plans are made in a number of stages, with earlier stages conditioning the latter ones. As such, travel planning is not simply sequential; the travel decision process can be iterative with the four phases repeating or even running in parallel for each decision such as destination, itinerary, things to do, and so on [25,57].

STTs play an increasingly important role in all the four phases of travel planning. It was predicted as early as the 1990s that the Internet would be accepted as the most important and effective tool to search travel information from the planning phase of travel [11]. Ho et al. [35] propose a conceptual framework of travelers' engagement in tourism information search and planning to understand their online and offline behavior and find previous knowledge and experience as the key basis for online travel information search. Xiang et al. [93] find users' perceived usefulness of the Internet for all the categories of travel decisions

such as where to visit, what to do, and where to stay has increased from 2007 to 2012. With the advent of new media such as social networks and smartphones, the focus of information search for travel decisions has been gradually shifting from only primary products (e.g., flights, lodging, etc.) to information to improve travel experience [17,88,91,93]. It is also noted, however, that travelers are concerned with the possible security or privacy breach with the increasing use of STTs [73].

It is generally believed that certain underlying characteristics of STTs drive their higher adoption and usage in travel planning. Xiang et al. [93] posit that the increased popularity of the Internet for trip decisions is because the Internet offers higher quality information and richer experience than ever. In the case of social media, the perceived value in travel information search goes up with users' perception of the information reliability and enjoyment [17]. In addition, tourists have found smartphones to be useful in helping them visit more places, have richer experience, and enjoy higher satisfaction with their trips overall [88]. No and Kim [60] further theorize that the four attributes of all the online tourism information sources – accessibility, information reliability, interaction, and personalization – are key to travel planning (with security and privacy as a potential barrier). However, although the factors of STTs leading to their use and usefulness are well studied, the underlying mechanism of how they are used successfully and/or satisfactorily for travel planning has not been examined. For this, we turn to the theory of exploration and exploitation.

### 2.3. Exploration and exploitation use

“Exploration of new possibilities” and “exploitation of existing capabilities” have long been recognized as two distinct mechanisms governing the learning and adaptive processes. March [56] points out that exploration refers to concepts as search, experimentation, discovery, and innovation, while exploitation is often associated with terms such as refinement, efficiency, implementation, and execution. Essence of exploration is the development of and experimentation with new alternatives, and exploitation focuses on improving existing resources and processes continuously [48]. Altogether, activities attributed to exploration and exploitation also constitute the essential course of adoption and use of information technology and systems [14,43].

The concepts of exploration and exploitation are often applied in the context of organizational learning. Exploration embodies the pursuit and development of new knowledge and skills and involves activities such as testing potential breakthrough ideas, experimenting with new skills, and discovering novel use of resources [3,33,80]. Although such actions can bring about the discovery of fundamentally innovative knowledge, the results of exploration can be “systematically less certain, more remote in time, and organizationally more distant from the locus of action and adaption” [56]. Exploitation, however, focuses on refining and extending the existing competences, capabilities, and resources [48,56]. Exploitation learning involves the use of and improvement on skills already known, and its learning process often calls for local search, experiential refinement, and selection and reuse of current routines [8]. This emphasis on improvement and refinement usually leads to fast learning results and gradual increase in efficiency [24].

The conceptualization of exploration and exploitation has been extended from organizational to individual learning and decision-making. Such extension has found strong support in a functional magnetic resonance imaging (fMRI) study, in which it is found that, when subjects take on exploitation tasks, there is stronger activation of the reward-related brain region, and that exploration tasks are more likely to activate the region of a subject's brain that governs the assessment of uncertainty and cognitive control [45].

For individuals, exploitation is defined as a behavior that optimizes the performance in the current tasks, while exploration involves a deviation from existing tasks and a quest for alternatives [2]. Many contrasts in characteristics between exploration and exploitation organizational learning, such as slow vs. fast and experimenting vs. refining, also exhibit at the individual level. Burton-Jones and Straub [14] suggest that in information system adoption, exploitation is related to the execution of routines and existing knowledge, whereas exploration involves searching for new and different uses of the systems. In the case of smartphones, it is found that user competence and perceived usefulness are positively correlated with both exploitative and explorative use, while perceived ease of use is significantly correlated with exploitative use but not with explorative use [43].

For this study, we conceptualize travelers' deployment of STT for trip planning as a combination of exploration and exploitation use. Considering the evolving nature of STTs, this study adopts the conceptual framework of exploration and exploitation in travel planning. We argue that although the use of STTs is involved in every stage of travel planning – idea formation, searching for information, evaluating alternatives for final decision, and booking – individuals interact with STTs differently at different travel planning stages, and such differences lead to differentiated consequences. Using STTs to perform the tasks of idea formation and searching for information in the first two phases of the travel decision-making process represents explorative use. Such use involves mostly unstructured tasks that require searching, testing, and discovery, and takes time to learn and adapt. For instance, when travelers simply want to get away but have not decided where to go and what to do, STTs can be used to explore the travel possibilities (phase one of travel planning) and search and compare different options (travel planning phase two). Indeed, it is found that finding “potential destinations to visit” and “stores or other places to shop” on the Internet garners moderate but increasing attention by travelers [93]. When successful, such ERU would likely enhance the overall experience of travelers. In addition, using STTs to complete the tasks of evaluating alternatives and booking of travel products in the last two phases of travel planning, which has been well adopted by travelers, represents exploitative use. Xiang et al. [93] find that more than two-thirds of the travelers surveyed in 2007–2012 used Internet for searching “information about a particular destination,” “hotel prices or places to stay,” and “airline fares and schedule/flight time.” These activities, involving routine tasks such as selecting vendors, comparing prices (both in travel planning phase three), and completing the purchase (phase four of the travel planning), aim at improving transaction efficiency and represent ETU for travel planning. With these two different styles of STT use that travelers engage in during the travel planning process, we present the research model in the next section.

### 2.4. Model and hypothesis development

Our study examines core antecedents that influence overall TES through both exploitative and explorative STT use. The research model is presented in Fig. 1. Consistent with the research on leisure satisfaction in travel and tourism [cf., 51,59,81], our model of travel satisfaction was subdivided into transaction satisfaction (TS) and TES, which correspond to the distinction between instrumental attributes (as being more cognitive oriented and thus derived from services and activities related to preparation and reservation) and expressive attributes (as being more emotion oriented or feeling oriented and thus derived from visual perspective and preference) of satisfaction. Because this study investigates the use of a specific category of information technology (i.e., the STTs), we include the factors that identify the STTs and influence the two different STT

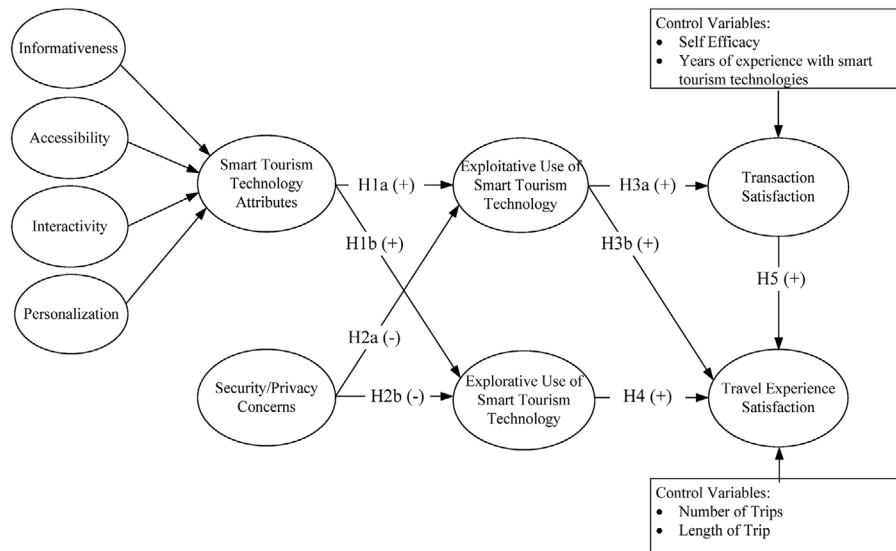


Fig. 1. Research Model.

uses. On the basis of the previous literature, key STTAs and security/privacy concerns are adopted as core antecedents of exploitive and ERU [60]. In the model, we posit that the former positively promotes the STT use and the latter rather inhibits the STT use. The two modes of STT use are simultaneously included in the model because both are involved in a complete travel planning process. We also argue that the explorative and exploitative use of STTs contributes to the TS and overall TES, respectively. We consider the explorative and exploitative use of STTs as parallel constructs because the travel planning phases are iterative, not sequential (see Section 2.2). Detailed arguments that support these relationships are discussed below.

### 2.5. Attributes of STTs

STTs can be identified by four key attributes – informativeness, accessibility, interactivity, and personalization – that can enhance the usability and perceived usefulness of the technologies [60]. *Informativeness* is a combination of the information quality and trust of information as provided by online tourism information sources. *Accessibility* is the degree to which travelers can easily access and use online tourism information sources. *Interactivity* facilitates travelers' immediate actions when using STTs through real-time feedback and active communications. Finally, *personalization* represents the ability of a traveler to obtain specific information to suit his/her personal trip planning needs. In this study, we investigate how these multiple attributes collectively motivate different uses of STTs.

Informativeness is particularly important to ETU: to select and book travel products such as air tickets, hotels, and car rentals effectively, travelers need to understand what relevant information to find, where to get the best deals, and sometimes even how to start a conversation with vendors or other travelers on the Internet [92]. The degree to which travelers can access useful information (accessibility) is associated with usability of STTs [13,47,93], which enables users to complete transactions easily. In general, travelers are likely to use interactive sites because of the experience of cognitive and affective absorption [87]. STTs' interactivity can therefore promote use for finding and purchasing travel products. Personalization allows for tailor-made services to fit individual traveler's needs and thus increases the perception of service quality [55,60]. In addition, travelers are more likely to use STTs that permit personalized searches (for instance, a hotel

booking site that not only accept price ranges, dates, and locations but also nearby landmarks, service facilities, and types of fellow travelers as search variables) to look for options and book that better meet their needs. Therefore, we propose the following hypothesis:

**H1a.** STTAs are positively associated with the ETU for travel planning.

The quality and reliability of information have a positive effect on their perceived value [17], resulting in more use of the technology means [90]. It is therefore important for STTs to be informative to allow travelers to discover and explore trip options. STTs with high level of accessibility allows for perceived ease of use [13,47,93], which facilitates travelers' search for travel ideas (i.e., explorative use). The interactivity of STTs provides more relevant and credible information due to the active participation of users [94], helping users construct initial ideas about travel, and enabling efficient travel information search. Chung and Koo [17] have found that user enjoyment and absorptive experience from the interactivity of STTs such as social media lead to a higher level of perceived value and consequently more usage. Personalized travel planning using STTs allows for complex travel options [90] and increasingly important exploratory activities such as on-the-fly travel decisions [89]. Therefore, we propose the following hypothesis:

**H1b.** STTAs are positively associated with the ERU for travel planning.

A user's concern for security and privacy is a factor that can influence the use of all the information technologies, including STTs. The ability to protect user's privacy and to provide secure transactions is a key evaluation factor for travelers to choose the right web site for travel planning [65]. Although security and privacy concerns (SPCs) no longer inhibit the use of information technology, they are still listed as the number one reason for not using the Internet for travel planning [93]. Such concerns are likely to be significant in the case of exploitative use, because sensitive information, such as personal identification and credit card data, is transmitted in the last two stages of travel planning. Therefore, we propose the following hypothesis:

**H2a.** SPCs are negatively associated with the ETU for travel planning.



Security and privacy can also be an issue for the ERU in the first two phases of trip planning. Search history can be captured by search engines when browsing information for travel ideas and options, and access to certain informational websites requires registration and/or some level of personal information and preference. In such cases, users may be concerned with how secure the STTs are and the information that they share when exploring and searching, although such concerns may not be as high as when financial transactions are involved. Therefore, we propose the following hypothesis:

**H2b.** SPCs are negatively associated with the ERU for travel planning.

## 2.6. Use of STTs and travel satisfaction

As exploitation is generally associated with execution and refinement of routines, the exploitative use of information technologies tends to lead to short-term task performance and efficiency gains [14,49,56]. ETU in travel planning refers to the evaluation and purchase of travel products, and as such, the performance gains of such use are due to the incremental improvement over the traditional processes of searching and booking with online and mobile means [35]. Indeed, trip transactions have been a leading application of e-commerce, partially due to such ETU being better and more efficient in accomplishing travelers' planning needs [90]. In addition, travelers have found STTs very useful or essential in selecting destinations or products and completing transactions [93]. As trip planning is part of the travel experience, when travelers can make better decisions and complete smoother transactions with STTs in the last two stages of travel planning, it is likely that they are more satisfied with the whole trip experience, although such effect by the ETU on TES may be less direct or significant than on TS. Therefore, we propose the following hypotheses:

**H3a.** ETU for travel planning is positively related to TS.

**H3b.** ETU for travel planning is positively related to TES.

ERU in travel planning focuses on generating trip ideas and searching for relevant information. Although this type of innovative ways of using technologies is not likely to generate process-specific results, there are other benefits to be gained [56,80]. As the use of Internet for searching and booking the primary travel product has reached saturation, travelers increasingly find values in using the Internet to search for secondary products such as museum tickets and shopping detour to enhance the trip experience [92]. Other STTs such as social media and smartphones allow for a richer, deeper, and more interactive research on various aspects of a trip [32,88]; the result of such explorations may include an exotic destination, an off-the-beaten-path tour, or even an on-the-go itinerary, all of which can enhance the overall travel experience in a novel, nontraditional way. Note that as ERU does not involve in the e-commerce activities of searching for and booking travel products, it is not likely to directly influence the TS of a traveler. Therefore, we propose the following hypothesis:

**H4.** ERU for travel planning is positively related to TES.

The transactions of making reservations and bookings are part of the overall travel experience. As mentioned in the previous literature, service quality leads to a satisfaction of the service, which in turn influences the satisfaction of the overall experience of which the service is a part of [4,59,62]. Therefore, it is perceivable that a better experience in travel transactions would

lead to a better experience of the whole trip, and we argue for the following hypothesis:

**H5.** TS is positively related to TES.

## 3. Research method

### 3.1. Data collection

We administered a survey to one of the largest online travel clubs in Korea. To expedite the data collection process, we developed a web survey, which consisted of nine demographic items, 26 questions with a seven-point Likert scale, and 18 Boolean-type questions. A pilot test with 30 graduate students was performed to ensure that the questions were unambiguous and that there were no technical errors, which might impede data collection. After the pilot study, a small number of questions were refined to improve their clarity. To overcome the problem of electronic survey that prevents random sampling and nonresponse rate tracking due to the anonymous nature of the Internet, we followed Andrews et al.'s [1] sample and participant selection quality criteria that use random sampling within an artificially defined sampling frame. Specifically, a key word search of "travel club" was performed using one of the largest search engines in Korea. The list of travel clubs from the search results was adopted as a sampling frame. Subsequently, size; access; and activity criteria were used to further define the sampling population; from which one travel club; with (1) the highest number of registered members (5878 members when accessed); (2) open for public participation; and (3) the highest number of posts within the past 30 days (i.e.; active club); was selected. We did so to ensure that the travel club we picked reflected the sampling population that fits well with the study context of smart tourism. Subsequently, with the support of the club administrator; an invitation to participate in the survey was posted as a notice on the general notice section. Two follow-up "reminder" invitations were then posted 1 week apart. A total of 325 responses were received; of which 319 responses were valid and thus used for further analysis. This represented a 5.43% response rate. Although this response rate is low; we were satisfied with the results because (1) the average ratio of the traffic in past 30 days to the total number of the registered members to the club was just little >20%; (2) no financial incentive was offered to the respondents for completing the survey; and (3) the club members surveyed did not know the researchers. The only incentive offered for completing the survey was that the club as a whole would receive a summary project report. The participants were assured that the results would only be reported in an aggregated form to guarantee their anonymity and confidentiality [69].

A summary of the demographic characteristics of 319 responses is provided in Table 1. Approximately 72.1% of the respondents in the sample have more than 1 year of experience in using STTs, and approximately 98% of the respondents in the sample make one or more trips per year, indicating a good representation of STT user community. Furthermore, the responses were based on a past trip with an average duration of 6.6 days. Approximately 39.5% of respondents in the sample spent 1 or 2 weeks for planning the trip.

### 3.2. Construct operationalization

All constructs in the survey were measured using multi-item scales. A conscientious effort was made to adapt existing measures validated from the previous studies for the latent constructs in this research. The specific items used in this study are shown in Appendix A.

**Table 1**  
Demographic Characteristics of Respondents (N = 319).

| Characteristics                                    |                       | Frequency | Percent |
|----------------------------------------------------|-----------------------|-----------|---------|
| Days of Trip Planning                              | less than one day     | 43        | 13.5%   |
|                                                    | 2–3 days              | 97        | 30.4%   |
|                                                    | 4–7 days              | 82        | 25.7%   |
|                                                    | 8–14 days             | 44        | 13.8%   |
|                                                    | 15–30 days            | 30        | 9.4%    |
|                                                    | more than 30 days     | 23        | 7.2%    |
| Number of leisure trip per year                    | never                 | 5         | 1.6%    |
|                                                    | 1–2                   | 186       | 58.3%   |
|                                                    | 3–4                   | 84        | 26.3%   |
|                                                    | 5 or more             | 44        | 13.8%   |
| Years of smart tourism technology use              | less than 1 year      | 89        | 27.9%   |
|                                                    | 1 years–2 years       | 82        | 25.7%   |
|                                                    | 2 years–3 years       | 45        | 14.1%   |
|                                                    | More than 3 years     | 103       | 32.3%   |
| Education level                                    | Middle school or less | 8         | 2.5%    |
|                                                    | High school           | 41        | 12.9%   |
|                                                    | College/University    | 164       | 51.4%   |
|                                                    | Graduate school       | 106       | 33.2%   |
| Sex                                                | Male                  | 172       | 53.9%   |
|                                                    | Female                | 147       | 46.1%   |
| Average duration of the trip chosen for the survey |                       | 6.6 days  |         |

### 3.3. Antecedents to the ETU and ERU

Antecedents (i.e., STT attributes and security/privacy concern constructs) to both ETU and ERU were measured with seven-point Likert rating systems, ranging from “1” being strongly disagree to “7” being strongly agree. Following the literature survey studies, on the attributes or properties of online tourism information sources [60,65], we adopted the five most widely used factors: (1) informativeness, (2) accessibility, (3) interactivity, (4) personalization, and (5) security and privacy. Although the tourism literature suggests that the first four attributes of STTs would have the positive effect on the tourism overall, the security and privacy issue in the tourism literature is identified as risk or nuisance of online channels. Thus, we drew a line between these two seemingly opposite characteristics of the attributes to bring differential effects to ETU and ERU. They were operationalized as follows.

Following Pavlou et al. [66], we define informativeness in this study as the degree to which information respondents (i.e., travelers) perceive STTs as useful, helpful, or relevant. Adapted from Luo [54] and Pavlou et al.’s [66], three items in this study are used to measure a perceptual construct [15].

Accessibility in the nontourism literature refers to the extent to which customers can easily find the information and/or services provided by online information resources [34]. As this attribute centers on the degree to which the information and/or services offered by an STT can be obtained and used without difficulty, three measures were adopted from No and Kim [60] to capture the extent to which respondents (i.e., travelers) can easily access and use STTs.

Ho and Lee [34], in their study of developing the scale for e-travel service quality, defined personalization as the extent to which the customized content of the website can provide a user with relevant and up-to-date information that meet his/her specific needs. Three items were adapted from No and Kim [60] and Pavlou et al. [66] to measure the extent to which respondents

(i.e., travelers) can obtain specific or perfect information that meet their needs from STTs.

Following the tourism literature, we define interactivity as an attribute that facilitates immediate actions such as real-time feedback and active communication, that is, information or idea sharing between users as well as administrators of STT. We adopted three measures from No and Kim [60] to measure interactivity.

Unlike the previous four attributes, information of security and privacy is a concern with STTs when the travelers use them. SPCs are defined as the travelers’ beliefs about a sellers’ *inability* and *unwillingness* to safeguard their monetary information from security breaches during transmission and storage [72] and to protect their personal information from improper use, disclosure to third parties, and secondary use without the travelers’ consent [76]. Three items were adapted from Pavlou et al. [66].

### 3.4. Exploitative and explorative use of STTs

Although these two constructs in the survey were measured using multi-items, they were operationalized with a different response format – Yes or No – to capture the breadth of both ETU and ERU. In doing so, we follow the procedural guide by Podsakoff et al. [68] for mitigating the common method bias. By using a different response format, instead of a typical Likert scale, would provide the respondents with a psychological separation between predictor and criterion variables when working with the questionnaire, and thus increase the methodological rigor. To this end, we adopted that of Rothaermel and Alexandre [71] to measure ETU and ERU by calculating the percentage of actual use versus possible use. That is,  $ETU = ((\text{actual}/\text{total}) \text{ smart travel exploitation})$ , and  $ERU = ((\text{actual}/\text{total}) \text{ smart travel exploration})$ .

Seven items were used to capture the breadth and frequency of ETU. These items cover a variety of ways that the STTs were used for activities associated with the last two stages of travel planning such as comparison shopping, reservation, and payment (see

Section 2.3). Although such activities are involved in all travel-related products, to avoid distraction of the subjects, the survey items focused on transportation, lodging, and dining, which together make up the bulk of travel-related transactions.

Eleven items were used to capture ERU, which corresponds to active search and discovery of information in both pre-trip travel planning and on-the-go guidance. As such, these measures are to capture a collective use of STTs in the first two stages of travel planning, such as when choosing a destination to visit [27,92] and building expectations for the upcoming trip [31]. In addition, because mobile- and location-based apps allow for real-time and on-the-go information search and exploration that lead to possible travel ideas and options [88], the measures are also to capture the ERU with smartphones and other mobile devices that may enrich touristic experience.

### 3.5. Transaction satisfaction and travel experience satisfaction

Consistent with the research in travel and tourism [cf.,51,59,81], the distinction between the travel TS (as being instrumental attribute) and the TES (as being expressive attribute) was maintained in our model of satisfaction. Four items for TS and five items for TES were adopted from Neal et al. [59]. We operationalized both constructs following the method used by Baker and Crompton [4]. Specifically, we used a subjective disconfirmation measure [84] for both transaction and TES requiring respondents to assess perceptions of satisfaction directly against their desired satisfaction level (defined as the level of satisfaction a traveler desires for the trip) and to record their evaluation with a single score. Instead of using the psychometric measure subject to the criticism of needing to go through the process of subtracting one measurement (expectations) from another measurement (perceptions) to compute the actual satisfaction [c.f.,10,20,53,82], we resorted to directly measure a respondent's perception of the level of satisfaction against an expectation standard.

To control for other contextual variables, which could also influence the TS, we included the following two control variables in our model: years of experience with STTs and self-efficacy. We also included the number of trips and the length of trip in the model to control their possible influence on the TES. The respondents were asked to indicate the duration of the trip they chose for this study. The number of trips per year was measured by four different classes. Years of experience with STTs was measured in four different levels, while self-efficacy was measured with three items using seven-point Likert scale as shown in Appendix A.

## 4. Analysis and results

The partial least square (PLS) method implemented in SmartPLS was chosen to perform a simultaneous evaluation of both quality of measurement (the measurement model) and hypothesized relationships (the structural model). The PLS technique is appropriate and well-suited for this study because it allows for latent constructs to be modeled with formative indicators in the structural model. This is an important requirement for our case as we model the second-order STTA construct as formative with four first-order constructs in this study [16]. We do this because we have conceptualized the STTAs as the collective characteristics or properties of online STTs as well as the collective support for the travel planning before and during the trip, which are captured through the first-order constructs in this study. Moreover, PLS is adequate for this study, in that it is more prediction oriented, which is suitable for assessing theories in the early stages of development [28]. Thus, it can be used to analyze the data as this

study is an initial attempt to theorize an atypical case of individual tourist's style of STT use in a unique setting of the tourism industry.

### 4.1. Measurement properties of variables

The measurement model was assessed through tests of convergent validity, discriminant validity, and reliability. First, exploratory factor analysis (EFA) was performed to check the potential latent factors in the development of measures. We found eight factors that have >0.7 factor loadings. Confirmatory factor analysis was consistent with EFA that showed no substantial cross-loading between the factors (see Table 2). These results support the unidimensionality of the scales in question. Internal consistency was assessed by using Cronbach's alpha. All the constructs used in this study are >0.75, showing strong reliability [61]. Subsequently, we evaluated convergent validity by testing composite reliability and average variance extracted (AVE) from the measures [40]. As shown in Table 2, composite reliability values were >0.85, Cronbach's alpha was >0.75, and AVE ranged from 0.65 to 0.89. These results indicate that the measurement model has high internal consistency and that convergent validity is confirmed [29]. The discriminant validity is established when (1) the square root of AVE for each construct is greater than the levels of correlation; (2) correlation between pairs of constructs is <0.9; and (3) cross-loadings of all items have a higher value in the defined construct than in any other constructs. The results in Tables 2 and 3 confirm the discriminant validity.

### 4.2. STTAs as the second-order construct

We theorize the STTA construct as a second-order construct with four first-order constructs representing characteristics of STTs. These multiple dimensions are grouped under the same multidimensional construct because each characteristic represents a distinct aspect of the overall latent construct of STTAs [46]. Specifically, following the suggestion by Jarvis et al. and Petter et al. [37,67], we developed a multidimensional construct that has a formative relationship between the second-order construct and the first-order constructs because multiple first-order constructs and their measurement items are necessary to fully capture the entire domain of the STTA construct.

One approach to justify the formative multidimensional constructs as opposed to the reflective construct is to contrast them with a unidimensional construct [37,67]. Reflective constructs should, by definition, be unidimensional, in that all the measurement items are measuring the same aspect of the unobservable construct, while multidimensional constructs are capturing multiple dimensions. To assess dimensionality of the STTA construct, we considered two separate models comparing alternative unidimensional and two-dimensional measurement models. Model 1 hypothesizes that a unidimensional factor accounts for the variance among all measurement items of the STTA construct. Model 2 hypothesizes that the measurement items divide into four uncorrelated first-order factors. Comparison of Model 1 ( $\chi^2 = 734.6$ , d.f. = 54) and Model 2 ( $\chi^2 = 121.3$ , d.f. = 54) indicates that the latter is a better-fitting model because it holds lower chi-square for the same degrees of freedom. This result supports for multidimensionality of the STTA construct. Moreover, a factor analysis was performed to evaluate the STTA construct as a first-order unidimensional construct by including all the items from each first-order construct as a single reflective construct. The result supported that the items making up the STTA construct were actually measuring four different aspects of the construct. Thus, the multidimensionality of the STTA construct appeared to be statistically supported.

**Table 2**  
Factor and cross-loadings (N = 319).

|                | Years on STTs | # of Trips   | Length of Trip | ERU          | ETU          | A            | SE           | INFO         | INT          | P            | SPC          | TES          | TS           | AVE   | Composite Reliability | Cronbach's Alpha |
|----------------|---------------|--------------|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------|-----------------------|------------------|
| Years on STTs  | <b>1.000</b>  | 0.321        | 0.041          | 0.024        | 0.050        | 0.026        | 0.130        | −0.008       | 0.065        | −0.030       | −0.032       | 0.033        | 0.034        | 1     | 1                     | 1                |
| # of Trips     | 0.321         | <b>1.000</b> | −0.004         | 0.079        | 0.063        | 0.033        | 0.048        | 0.061        | 0.082        | 0.000        | 0.002        | 0.054        | 0.136        | 1     | 1                     | 1                |
| Length of Trip | 0.041         | −0.004       | <b>1.000</b>   | 0.023        | 0.006        | 0.102        | 0.102        | 0.202        | 0.229        | 0.097        | −0.128       | 0.131        | 0.197        | 1     | 1                     | 1                |
| ERU            | 0.024         | 0.079        | 0.023          | <b>1.000</b> | 0.521        | 0.238        | 0.165        | 0.258        | 0.218        | 0.287        | −0.345       | 0.140        | 0.176        | 1     | 1                     | 1                |
| ETU            | 0.050         | 0.063        | 0.006          | 0.521        | <b>1.000</b> | 0.240        | 0.176        | 0.192        | 0.132        | 0.258        | −0.392       | 0.089        | 0.139        | 1     | 1                     | 1                |
| A1             | −0.004        | 0.044        | 0.008          | 0.202        | 0.210        | <b>0.883</b> | 0.302        | 0.536        | 0.224        | 0.392        | −0.181       | 0.347        | 0.442        | 0.818 | 0.931                 | 0.889            |
| A2             | 0.042         | 0.041        | 0.068          | 0.240        | 0.259        | <b>0.931</b> | 0.363        | 0.578        | 0.271        | 0.429        | −0.170       | 0.425        | 0.504        |       |                       |                  |
| A3             | 0.030         | 0.007        | 0.192          | 0.202        | 0.182        | <b>0.899</b> | 0.359        | 0.561        | 0.378        | 0.469        | −0.169       | 0.403        | 0.423        |       |                       |                  |
| SE1            | 0.099         | 0.034        | 0.105          | 0.176        | 0.202        | 0.362        | <b>0.934</b> | 0.326        | 0.185        | 0.216        | −0.049       | 0.235        | 0.259        | 0.868 | 0.952                 | 0.924            |
| SE2            | 0.107         | 0.037        | 0.048          | 0.160        | 0.169        | 0.341        | <b>0.952</b> | 0.317        | 0.168        | 0.209        | −0.031       | 0.268        | 0.274        |       |                       |                  |
| SE3            | 0.153         | 0.061        | 0.131          | 0.128        | 0.124        | 0.355        | <b>0.909</b> | 0.323        | 0.144        | 0.193        | −0.034       | 0.265        | 0.294        |       |                       |                  |
| INFO1          | −0.002        | 0.082        | 0.166          | 0.230        | 0.126        | 0.522        | 0.267        | <b>0.879</b> | 0.295        | 0.395        | −0.168       | 0.392        | 0.3478       | 0.794 | 0.920                 | 0.870            |
| INFO2          | 0.019         | 0.063        | 0.196          | 0.221        | 0.143        | 0.554        | 0.356        | <b>0.912</b> | 0.308        | 0.421        | −0.102       | 0.484        | 0.451        |       |                       |                  |
| INFO3          | −0.039        | 0.019        | 0.178          | 0.237        | 0.242        | 0.569        | 0.298        | <b>0.882</b> | 0.306        | 0.463        | −0.216       | 0.470        | 0.445        |       |                       |                  |
| INT1           | 0.090         | 0.100        | 0.138          | 0.143        | 0.069        | 0.189        | 0.087        | 0.138        | <b>0.751</b> | 0.344        | 0.117        | 0.181        | 0.213        | 0.666 | 0.857                 | 0.755            |
| INT2           | 0.016         | −0.018       | 0.214          | 0.182        | 0.097        | 0.202        | 0.167        | 0.251        | <b>0.824</b> | 0.407        | −0.022       | 0.242        | 0.260        |       |                       |                  |
| INT3           | 0.062         | 0.114        | 0.200          | 0.200        | 0.142        | 0.365        | 0.166        | 0.388        | <b>0.870</b> | 0.423        | −0.092       | 0.267        | 0.350        |       |                       |                  |
| P1             | −0.072        | −0.005       | 0.140          | 0.244        | 0.208        | 0.475        | 0.191        | 0.492        | 0.479        | <b>0.889</b> | −0.164       | 0.346        | 0.396        | 0.763 | 0.906                 | 0.845            |
| P2             | 0.051         | −0.004       | 0.134          | 0.211        | 0.192        | 0.373        | 0.159        | 0.412        | 0.469        | <b>0.873</b> | −0.180       | 0.323        | 0.330        |       |                       |                  |
| P3             | −0.054        | 0.009        | −0.031         | 0.302        | 0.284        | 0.395        | 0.230        | 0.342        | 0.382        | <b>0.858</b> | −0.205       | 0.280        | 0.348        |       |                       |                  |
| SPC1           | −0.025        | −0.010       | −0.092         | −0.332       | −0.367       | −0.152       | −0.004       | −0.155       | −0.021       | −0.227       | <b>0.912</b> | −0.112       | −0.056       | 0.895 | 0.972                 | 0.961            |
| SPC2           | −0.035        | 0.000        | −0.126         | −0.293       | −0.340       | −0.165       | −0.033       | −0.154       | −0.015       | −0.179       | <b>0.956</b> | −0.117       | −0.062       |       |                       |                  |
| SPC3           | −0.036        | 0.011        | −0.129         | −0.333       | −0.381       | −0.203       | −0.056       | −0.203       | −0.020       | −0.193       | <b>0.963</b> | −0.116       | −0.084       |       |                       |                  |
| SPC4           | −0.024        | 0.008        | −0.135         | −0.342       | −0.393       | −0.200       | −0.058       | −0.175       | −0.019       | −0.188       | <b>0.955</b> | −0.128       | −0.096       |       |                       |                  |
| TES1           | 0.074         | 0.001        | 0.042          | 0.085        | 0.066        | 0.371        | 0.213        | 0.401        | 0.189        | 0.267        | −0.145       | <b>0.807</b> | 0.322        | 0.655 | 0.905                 | 0.868            |
| TES2           | 0.019         | 0.004        | 0.115          | 0.063        | 0.043        | 0.293        | 0.164        | 0.377        | 0.252        | 0.224        | −0.057       | <b>0.735</b> | 0.411        |       |                       |                  |
| TES3           | 0.042         | 0.033        | 0.106          | 0.152        | 0.076        | 0.307        | 0.256        | 0.374        | 0.246        | 0.358        | −0.112       | <b>0.852</b> | 0.449        |       |                       |                  |
| TES4           | 0.007         | 0.115        | 0.131          | 0.150        | 0.098        | 0.432        | 0.213        | 0.470        | 0.258        | 0.280        | −0.100       | <b>0.847</b> | 0.433        |       |                       |                  |
| TES5           | −0.021        | 0.074        | 0.148          | 0.119        | 0.079        | 0.351        | 0.270        | 0.424        | 0.225        | 0.343        | −0.081       | <b>0.800</b> | 0.414        |       |                       |                  |
| TS1            | 0.039         | 0.109        | 0.218          | 0.166        | 0.168        | 0.451        | 0.276        | 0.401        | 0.374        | 0.351        | −0.108       | 0.476        | <b>0.832</b> | 0.656 | 0.884                 | 0.825            |
| TS2            | 0.031         | 0.107        | 0.158          | 0.212        | 0.172        | 0.395        | 0.261        | 0.462        | 0.295        | 0.388        | −0.053       | 0.470        | <b>0.837</b> |       |                       |                  |
| TS3            | −0.008        | 0.067        | 0.091          | 0.044        | 0.015        | 0.437        | 0.176        | 0.483        | 0.208        | 0.332        | −0.064       | 0.409        | <b>0.783</b> |       |                       |                  |
| TS4            | 0.043         | 0.154        | 0.161          | 0.129        | 0.076        | 0.352        | 0.238        | 0.433        | 0.231        | 0.258        | −0.029       | 0.438        | <b>0.786</b> |       |                       |                  |

Note: A = Accessibility, ERU = Explorative Use of Smart Tourism Technologies, EXU = Exploitative Use of Smart Tourism Technologies, SE = Self-Efficacy, INFO = Informativeness, INT = Interactivity, P = Personalization, SPC = Security/Privacy Concerns, TES = Travel Experience Satisfaction, TS = Transaction Satisfaction, AVE = Average Variance Extracted.

Of the four types of second-order models that combine formative and reflective measures [37], this study applies Type II because the reflective indicators combine for first-order constructs, which cause a formative second-order construct. Therefore, we modeled the STTA empirically as a formative second-order construct that consists of informativeness, accessibility, interactivity, and personalization, each of which represents facets of STTAs that could be separate constructs but remain integral parts of STTA at a more abstract level. Although Jarvis et al. [37] explain decision rules for determining whether a construct is reflective or formative at the item level, we extended its

application to the construct level to assess the adequacy of the formative second-order construct we adopted to capture a substantive level of the STTA.

First, we consider the theoretical direction of causality between each construct and its second-order construct. To verify this decision rule, Jarvis et al. [37] ask the researcher to consider the nature of the measures. We believe it makes more sense to say that travelers would perceive the benefits of STTs because they feel informed, accessed, interactive, and personalized response when using the STTs than to say that they perceive enhanced information quality, access, interaction, and personalized response because

**Table 3**  
Construct Correlations.

|                | Years on STTs | # of Trips | Length of Trip | STTA         | ERU      | ETU      | SE           | SPC          | TES          | TS           |
|----------------|---------------|------------|----------------|--------------|----------|----------|--------------|--------------|--------------|--------------|
| Years on STTs  | <b>1</b>      |            |                |              |          |          |              |              |              |              |
| # of Trips     | 0.321         | <b>1</b>   |                |              |          |          |              |              |              |              |
| Length of Trip | 0.041         | −0.004     | <b>1</b>       |              |          |          |              |              |              |              |
| STTA           | 0.011         | 0.051      | 0.188          | <b>0.691</b> |          |          |              |              |              |              |
| ERU            | 0.024         | 0.079      | 0.023          | 0.317        | <b>1</b> |          |              |              |              |              |
| ETU            | 0.050         | 0.063      | 0.006          | 0.267        | 0.721    | <b>1</b> |              |              |              |              |
| SE             | 0.130         | 0.048      | 0.102          | 0.367        | 0.165    | 0.176    | <b>0.932</b> |              |              |              |
| SPC            | −0.032        | 0.002      | −0.128         | −0.204       | −0.345   | −0.392   | −0.040       | <b>0.946</b> |              |              |
| TES            | 0.033         | 0.054      | 0.131          | 0.513        | 0.140    | 0.089    | 0.276        | −0.125       | <b>0.809</b> |              |
| TS             | 0.034         | 0.136      | 0.197          | 0.593        | 0.176    | 0.139    | 0.297        | −0.079       | 0.678        | <b>0.810</b> |

Note: STTA = Smart Tourism Technology Attributes, SE = Self-Efficacy, SPC = Security/Privacy Concerns, TES = Travel Experience Satisfaction, TS = Transaction Satisfaction, ERU = Explorative Use of Smart Tourism Technology, ETU = Exploitative Use of Smart Tourism Technology. Bolded diagonal elements are the square root of average variance extracted (AVE).



they know the attributes of STTs. Clearly, informativeness, accessibility, interactivity, and personalization are identified as distinct characteristics of STTs that, together, share the ingredient of being STTA.

The second decision rule we consider was to examine the interchangeability of the first-order constructs. According to this rule, the multiple first-order constructs under the second-order construct are not interchangeable in the formative representation of the second-order construct. For instance, looking at the formative version of the STTA construct, we can see that there are four different aspects being measured. Tourism literature commonly identifies four characteristics as different representations of the STTAs. Removing one of these construct changes how we understand and interpret the STTA.

Similarly, another decision rule to identify formative constructs asks if the measures of the construct have the same antecedents and consequences. Formative constructs are composites or indices that are made up of measures that may be very different; thus, it is not necessary for the measures to have the same antecedents and consequences. For example, with the STTA, the reasons (or antecedents) for enhanced informativeness on the travel-related websites or apps may be different from the reasons for enhanced accessibility. These items may share some of the antecedents and/or consequences, but it is not required. The assumption for the reflective operationalization of the STTA, however, is that the same factor caused a change in the STTA construct, which is then reflected in each of the subconstructs. We theorized in the current study that the different usage styles driven with types of task involved in the travel planning process would result in differential satisfactions, which is aligned with the assumption of a formative construct. Overall, with both theoretical and empirical grounds, the conceptualization of the STTA as second-order construct appears justified.

#### 4.3. Common method bias analysis

Following the suggestion by Podsakoff et al. [68], the two primary ways to control for method biases were performed: (1) the design of the study's procedures and (2) statistical remedies.

First, we used different response formats between predictor and criterion variables to bring in techniques for controlling common method bias and thus minimize the possible influence on the response process. For example, the seven-point Likert scale was used to measure antecedents of the traveler's styles of STT use, whereas both ETU and ERU used the Boolean format (i.e., Yes or No). Doing it this way is known to provide the respondents with a psychological separation during the response, thereby reducing method biases caused by commonalities in scale end points and anchoring effects [68]. Furthermore, we adopted the subjective disconfirmation measure for both transaction and TES. With it, the

respondents were asked to assess perceptions of satisfaction directly against their desired satisfaction level (defined as the level of satisfaction a traveler desires for the trip). By avoiding the use of bipolar scale values and providing verbal labels for the midpoints of scales, we hoped to reduce acquiescence bias [cf. 83]. The format and instructions used for the subjective disconfirmation measure are reproduced in the following table (Table 4).

For statistical controls, two types of statistical analyses were conducted to assess the threat of common method bias: (1) Harman's one-factor test [68] and (2) Lindell and Whitney's [52] marker variable test. In Harman's one-factor test, the emergence of a single factor that accounts for a large proportion of the variance in factor analysis suggests a common method bias [68]. No such single factor emerged and the first factor accounted for only 21.04% of the total 72.96% variance. Second, the Lindell–Whitney [52] marker variable test uses a theoretically unrelated marker variable to adjust the correlations among the model's principal constructs. Because a marker variable does not have a theoretically expected relationship with the study's principal constructs, a high correlation would indicate common method bias. We used a three-item scale variable, need for cognition ( $\alpha = 0.78$ ), for which there exists little theoretical basis for a relationship with our research variables. The average correlation of the study's principal constructs with it ( $r = 0.091$ ) was low and insignificant, providing no evidence of common method bias. Taken together, we conclude that common method bias is not a serious threat in this study.

#### 4.4. Results of the structural model

Having validated the measurement model, the next step was testing the hypothesized relationships among latent constructs in the structural model. The assessment and estimation of the structural model were conducted using SmartPLS [70]. To determine the precision of estimation in this particular PLS method, a bootstrapping procedure with resampling of 200 subsamples was used to determine the statistical significance of the parameter estimates. On the basis of the results of this procedure, the structural model was assessed examining the magnitude, statistical significance of the path coefficients, and  $R^2$  in the structural model. Overall, the results suggest a satisfactory fit of the model to the data. The  $R^2$  values of the dependent constructs were 0.46 and 0.22 for TES and TS, respectively. In addition, the  $R^2$  values of the endogenous constructs as evaluated were 0.19 and 0.18 for ETU and ERU, respectively. The results of the PLS analysis are shown in Fig. 2.

We first note that ERU was significantly and positively associated with TES ( $\beta = 0.352$ ,  $p < 0.001$ ), supporting H4. The results of ETU were mixed. ETU was significantly and positively associated with the TS ( $\beta = 0.207$ ,  $p < 0.05$ ), supporting H3a. However, for TES, it showed statistically insignificant, albeit a

**Table 4**

Trip Satisfaction Measurement: For each of the following attributes, please circle the number that indicates how your trip's satisfaction compares with your desired satisfaction level.

| My satisfaction for the trip was: |                                          |   |   |                                           |   |                                           |   |
|-----------------------------------|------------------------------------------|---|---|-------------------------------------------|---|-------------------------------------------|---|
|                                   | Lower than my desired satisfaction level |   |   | The same as my desired satisfaction level |   | Higher than my desired satisfaction level |   |
| When it comes to . . . . .        |                                          |   |   |                                           |   |                                           |   |
| Service quality                   | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Comfort                           | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Safety                            | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Logistics and Itinerary           | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Enjoyment                         | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Value for the money               | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Excitement                        | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Memorability                      | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |
| Impression                        | 1                                        | 2 | 3 | 4                                         | 5 | 6                                         | 7 |

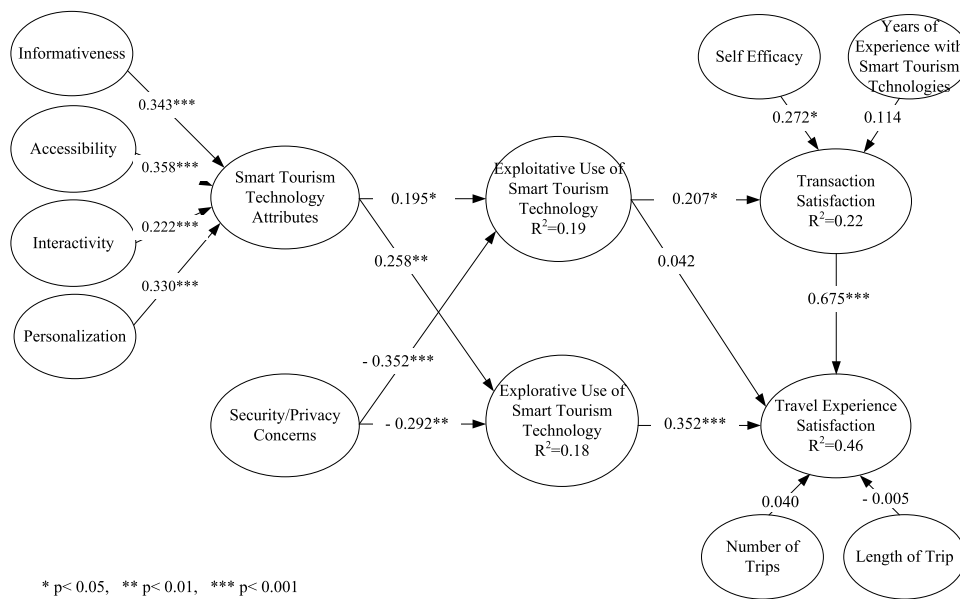


Fig. 2. Results of the structural model.

positive association with TES. Thus, H3b was not supported. The interpretation of these interesting results is discussed in the next section.

Second, the relationships between both styles of use and their antecedents turned out as expected. STTAs were significantly and positively associated with both ETU ( $\beta = 0.195$ ,  $p < 0.05$ ) and ERU ( $\beta = 0.258$ ,  $p < 0.01$ ), supporting H1a and H1b, respectively. As hypothesized, the effect of SPCs on both ETU and ERU turned out to be negative and significant, supporting H2a ( $\beta = -0.352$ ,  $p < 0.001$ ) and H2b ( $\beta = -0.292$ ,  $p = 0.01$ ). These results indicate that multitude properties of STTs have a differential effect on the traveler's consumption style of STTs before and during their trips. Specifically, while STTAs stimulate the travelers to engage in both explorative and exploitative use, the traveler's concerns of security and privacy with online tourism technologies hinder both styles of STT use. As expected and controlled, self-efficacy showed a significant and positive effect on the TS ( $\beta = 0.272$ ,  $p < 0.05$ ).

To ensure that our above statistical tests have adequate power, we performed a power analysis to calculate power values for our PLS model using Cohen's [19] power table for multiple regression analysis. As PLS estimates a structural model block by block, we calculated power values block by block [cf.,50]. Because the TS block involves control variables, we took a hierarchical approach to analyze power for incremental explanation of variance, as suggested by Cohen [19]. Thus, we conducted nine power analyses. The power values for all these analyses ranged from 0.88 to 0.96 indicating that our sample provides sufficient power at the 0.95 level to detect effects when they are present [19]. As the results are

validated, we will discuss the implications of this study in more detail in the next section.

## 5. Discussion and conclusion

Given the central role of STTs in travel planning, it is important to understand how travelers actually use STTs and the consequences of such usage. We explore the different types of STT use and have revealed several significant and interesting findings. First, our result indicates that explorative use is strongly related to TES ( $\beta = 0.352$ ,  $p < 0.001$ ). Knowing that such use is effective in idea formation and information search stages of travel planning, we can interpret this result in two ways. Because finding the right destinations, places to visit, and itinerary is a critical aspect of travel experience, using STTs for this purpose understandably has a major effect on the overall travel satisfaction. In addition, STTs allow travelers to explore many more alternatives in making trip planning decisions than do the traditional information sources such as printed brochures and travel agents. Such "uncertainty" enabled by ERU may actually add to the pleasure of travel because of the sense of adventure and exploration even in the planning phase [86]. Interestingly, we find that exploitative use, an important determinant of TS ( $\beta = 0.207$ ,  $p < 0.05$ ), only indirectly affects the overall travel experience through TS. Because the ETU is associated directly with the evaluation of travel products and booking, the last two stages of travel planning, the result of such use is therefore a more efficient and smoother transaction, which in turn contributes to the overall travel experience. These

Table 5  
Mediation Tests.

| Models  | Path A     | t-value | Path B    | t-value | Path C     | t-value | Soble test | p     |
|---------|------------|---------|-----------|---------|------------|---------|------------|-------|
| Model 1 | STTA → ETU | 1.7377  | ETU → TES | 0.176   | STTA → TES | 1.504   | 2.258      | 0.012 |
| Model 2 | STTA → ERU | 2.497   | ERU → TES | 5.596   | STTA → TES | 0.314   | 2.001      | 0.023 |
| Model 3 | SPC → ETU  | 4.034   | ETU → TES | 0.176   | SPC → TES  | 0.929   | -1.988     | 0.023 |
| Model 4 | SPC → ERU  | 2.982   | ERU → TES | 5.596   | SPC → TES  | 0.416   | -2.047     | 0.020 |

interesting findings led us to test medication effects of STTA and SPC on TES using Sobel's test [78].

To generate the required *t*-test values, we ran bootstrapping for four sets of independent PLS models. Models 1 and 2 included paths from the STTA variable to the two mediator variables (i.e., exploitative and explorative uses) and paths from the mediator variables to the TES. Models 3 and 4 included paths from the SPC variable to the two mediator variables (i.e., ETU and ERU) and paths from the mediator variables to the TES. The *t*-test statistics of each path on both models are shown in Table 5. On the basis of the Sobel's test results, we concluded that the effects of STTA as well as SPC on TES were significantly and fully mediated by these two mediating factors. These results of Sobel's test confirm the proposed mediating role of both ETU and ERU on TES with minimal direct relationships between STTA and TES and between SPC and TES.

Second, this study explores the role of STTAs in motivating the ETU and ERU. Although No and Kim [60] identify four factors relevant to the usefulness of STTs, how they actually facilitate STT use by travelers to achieve overall TES is not known. Our results show that the STTAs contribute to travelers' adoption of STTs in travel planning, and the influence is stronger on explorative use than on exploitative use. This may be explained by the fact that the travelers' exploitative use of STTs is already common and saturated [89], while the explorative use is still emerging and growing [92]; higher level of informativeness, accessibility, interactivity, and personalization can better motivate travelers to experiment with STTs in travel planning in innovative ways such as discovering exotic destinations and finding places to visit on the go. Although the results clearly indicate the pattern of relationships, one may be concerned with the strength of the identified pattern due to low  $R^2$  values of the endogenous constructs for the ETU and ERU (0.19 and 0.18, respectively). However, given the objective of the study, which is to test the relationships between core antecedents (STTA and SPC) and the different STT uses in extending the understanding of how the use of STT leads to overall TES, we believe that the focal interest is on determining whether underlying relationships between core antecedents (STTA and SPC) and the two modes of STT use exist and are statistically significant (with low *p*-value), as opposed to making strong prediction of how STT is used ( $R^2$  values). The high Cohen's power values for those relationships also indicate that our sample provides sufficient power to detect effects when they are present [19], thus signifying practical significance as well.

Third, this study shows that security and privacy are still a concern to travelers when using STTs in travel planning. SPCs have been examined as a critical factor that hinders online transactions or information disclosure [e.g.,21], but this study points out specifically that these concerns negatively affect both explorative and ETU in all the stages of travel planning. It is interesting to note that travelers are concerned with security and privacy even when they explore the online information channels in idea formation and information search, although the negative effects are not as strong as in the case of exploitative use, which often involves exchange of private information and financial transactions.

This study advances the theoretic development in the stream of STT research. As one of the few empirical research to examine explorative and exploitative use of information technology and the first to extend this framework to the tourism setting, this study investigates the influence of technology attributes on the types of STT use in travel planning and compares their effects on the overall travel experience. We show that exploitative use and explorative use do play important yet differentiated roles for the travelers, leading to distinctive types of satisfaction.

The current study also provides the context for understanding travel planning with STTs as an important foundation for sustainable services in travel and tourism. Although exploitative

use facilitates product comparison and booking transactions, ERU in travel planning enriches trip experience by enhancing information search, processing, and sharing, as well as enabling about the discovery of novel and unique travel opportunities. This study suggests that ERU in travel planning directs "tourists gaze" by fashioning and reinforcing travelers' anticipation to visit places [86] and by providing mental pleasure through stimulating fantasies and daydreams even before the trip begins [85]. Our findings offer several managerial implications. Internet, and later on social media and smartphone with the location-based service, has fundamentally changed the nature of travel and tourism communication and marketing [31]. STTs have fostered a new rich, socially created environment whereby the "army of travelers" is now the central player within the value structure [74]. As such, STTs pose a significant and growing challenge to marketers and managers alike because they must find novel ways to distinguish themselves and to stay competitive. Our results suggest that when designing a website or app for travel planning, it is important to pay attention to informativeness, accessibility, interactivity, and personalization, because these attributes will ultimately lead to overall satisfaction of travel experience. For instance, a site with active and customized user review, thus highly informative and personalizable, can help travelers obtain better trip experience and therefore attract more users to the site. A particular STT can also target explorative or exploitative use, thus different stages of travel planning, by emphasizing certain attributes. At the same time, this study illustrates that security and privacy issues are still important to users. So, even for STTs that are designed for tasks in travelers' early stages of trip planning – generating ideas, browsing information, searching for recommendations, etc. – clear provision of security and privacy can ease users' concerns and possibly increase usage. Adopting well-known certificate can also be effective in reducing SPCs.

As with all research, this study has limitations that must be considered. A potential limitation concerns the nature of the sample used in this analysis. Although the sample shows good representation of travelers, these findings can only be generalized to STT users within the sampling frame. Thus, no claim of external validity for this study's findings, particularly in cases of nonleisure travels, can be made. It is also worth noting that the sample was composed of Korean travelers, selected from the member of one of the largest online travel clubs in Korea. We anticipate that there could be systematic differences among countries in aspects such as type of STTs available or style of STT use that make generalization of the results to other settings difficult. We believe that a future study that compares the current findings to those in other countries – in particular the US – could yield insight into how the STT use in travel planning in other cultural and regulation settings. Another limitation is the cross-sectional design used in this research. Although the results provide strong support for the hypotheses, our model could benefit from a longitudinal research design to provide strong inferences for causality relationships for travel experience from a temporal perspective [30,39]. This is particularly useful because of the fast-changing nature of STTs. Future studies can consider designing a longitudinal study to assess the style over time as the combination of available STTs and their attributes evolve.

Despite the limitations described above, our study adds to the stream of literature on smart tourism and STTs. The goal of this paper is to identify the mechanism of the actual usage of STTs in travel planning and to understand how the different uses influence the travel experience, and the result shows the close relationships among STTAs, exploration and exploitation use, and travel satisfaction. Therefore, we believe that this study contributes to the advancement of our understanding of STTs in travel planning, and future extensions to this theme can be expected.

## Appendix A. Survey Items

| A Construct                            | B Items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interactivity <sup>a</sup>             | 1. A lot of other users' questions and answers can be found on tourism websites and apps.<br>2. The tourism websites and apps that I use are highly responsive to users.<br>3. It is easy to share content (e.g. tourism information) on tourism websites and apps.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Informativeness <sup>a</sup>           | 1. Tourism websites and apps provide useful information of the destination(s) and the trip.<br>2. I expect tourism websites and apps to help me evaluate the destination(s) and the trip.<br>3. The tourism websites and apps enable me to complete my travel with detailed information provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Accessibility <sup>a</sup>             | 1. I can use tourism websites and apps anytime and anywhere.<br>2. I find tourism websites and apps easy to use.<br>3. Tourism websites and apps can be easily found.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Personalization <sup>a</sup>           | 1. Tourism websites and apps allow me to receive tailored information.<br>2. I can interact with tourism websites and apps to get personalized information.<br>3. The personalized information provided by tourism websites and apps meets my needs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Security/Privacy Concerns <sup>a</sup> | 1. I am concerned that too much personal information is collected when I use tourism websites and apps.<br>2. I have doubts about how well my privacy is protected when I use tourism websites and apps.<br>3. My personal information could be subject to misuse and unauthorized access when transacting through tourism websites and apps.<br>4. I am concerned with the security of sensitive information when I use tourism websites and apps.                                                                                                                                                                                                                                                                                                                                                                                                |
| Self-Efficacy <sup>a</sup>             | 1. I have necessary skills to use tourism apps or websites.<br>2. I have knowledge of using tourism apps or websites.<br>3. I am confident of using tourism apps or websites even if there is no one around to show me how to do it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Smart Tourism Exploitation (YES/NO)    | 1. For this trip, I used travel booking or review websites to compare and select travel vendors for transportation, lodging, or dining.<br>2. For this trip, I watched online videos or checked personal blogs to compare and select travel vendors for transportation, lodging, or dining.<br>3. For this trip, I used social media to compare and select travel vendors for transportation, lodging, or dining.<br>4. For this trip, I used mobile apps to compare and select travel vendors for transportation, lodging, or dining.<br>5. For this trip, I used travel booking websites to complete booking for transportation, lodging, or dining.<br>6. For this trip, I used mobile apps to complete booking for transportation, lodging, or dining.<br>7. For this trip, I compared and/or booked pre-packaged travel deals offered online. |
| Smart Tourism Exploration (YES/NO)     | For this trip, I used travel booking or review websites to discover and determine destination(s) to go to and itinerary.<br>2. For this trip, I watched online videos or checked personal blogs to discover and determine destination(s) to go to and itinerary.<br>3. For this trip, I used social media to discover and determine destination(s) to go to and itinerary.<br>4. For this trip, I used mobile app to discover and determine destination(s) to go to and itinerary.<br>5. For this trip, I used travel booking or review websites to discover and determine attractions and activities at the destination(s).<br>6. For this trip, I watched online videos or checked personal blogs to discover and determine attractions and activities at the destination(s).                                                                    |

(Continued)

| A Construct                         | B Items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | 7. For this trip, I used social media to discover and determine attractions and activities at the destination(s).<br>8. For this trip, I used mobile app to discover and determine attractions and activities at the destination(s).<br>9. For this trip, I used smart, location-based mobile apps to discover attractions and activities at the destination(s).<br>10. For this trip, I used smart, location-based mobile apps to discover attractions and activities at the destination(s).<br>11. For this trip, I used smart, location-based mobile apps to select and determine attractions and activities for the destination(s). |
| Transaction Satisfaction            | My Satisfaction with transaction was (1 being "lower than my desired satisfaction level", 4 being "the same as my desired satisfaction level", and 7 being "Higher than my desired satisfaction level")<br>When it comes to . . . . .<br>1. Service quality<br>2. Comfort<br>3. Safety<br>4. Logistics and Itinerary                                                                                                                                                                                                                                                                                                                    |
| Travel Experience Satisfaction      | My satisfaction with Travel Experiences was (1 being "lower than my desired satisfaction level", 4 being "the same as my desired satisfaction level", and 7 being "Higher than my desired satisfaction level")<br>When it comes to . . . . .<br>1. Enjoyment<br>2. Value for the money<br>3. Excitement<br>4. Memorability<br>5. Impression                                                                                                                                                                                                                                                                                             |
| Years on Smart Tourism Technologies | Total number of years using smart tourism technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

<sup>a</sup>Seven point Likert scale (1 being strongly disagree and 7 being strongly agree).

## References

- [1] D. Andrews, B. Nonnecke, J. Preece, Electronic survey methodology: a case study in reaching hard-to-involve internet users, *Int. J. Hum. Comput. Interact.* 6 (2) (2003) 185–210.
- [2] G. Aston-Jones, J.D. Cohen, Adaptive gain and the role of the locus coeruleus–norepinephrine system in optimal performance, *J. Comp. Neurol.* 493 (1) (2005) 99–110.
- [3] S. Auh, B. Menguc, Balancing exploration and exploitation: the moderating role of competitive intensity, *J. Bus. Res.* 8 (12) (2005) 1652–1661.
- [4] D.A. Baker, J.L. Crompton, Quality, satisfaction and behavioral intentions, *Ann. Tour. Res.* 27 (3) (2000) 785–804.
- [5] S. Banerjee, A.Y.K. Chua, In search of patterns among travellers' hotel ratings in Tripadvisor, *Tour. Manage.* 53 (2016) 125.
- [6] B. Bargeman, H. van der Poel, The role of routines in the vacation decision-making process of dutch vacationers, *Tour. Manage.* 7 (4) (2006) 707–720.
- [7] U. Bastida, T.-C. Huan, Performance evaluation of tourism websites' information quality of four global destination brands: Beijing, Hong Kong, Shanghai, and Taipei, *J. Bus. Res.* 67 (2) (2014) 167–170.
- [8] J.A.C. Baum, S.X. Li, J.M. Usher, Making the next move: how experiential and vicarious learning shape the locations of chains' acquisition, *Adm. Sci. Q.* 5 (4) (2000) 766–801.
- [9] P. Björk, H. Kauppinen-Räsänen, Contemporary insights to the dynamic pre-trip information sourcing behaviour, *Tour. Hosp. Res.* 5 (1) (2015) 39–53.
- [10] T.J. Brown, G.A. Churchill, J.P. Peter, Improving the measurement of service quality, *J. Retail.* 9 (1) (1993) 127–139.
- [11] D. Buhalis, Strategic use of information technologies in the tourism industry, *Tour. Manage.* 9 (5) (1998) 409–421.
- [12] D. Buhalis, M. Foerste, Socomo marketing for travel and tourism: empowering co-creation of value, *J. Destin. Market. Manage.* 4 (3) (2015) 151–161.
- [13] D. Buhalis, R. Law, Progress in information technology and tourism management: 20 years on and 10 years after the internet—the State of Etourism Research, *Tour. Manage.* 9 (4) (2008) 609–623.
- [14] A. Burton-Jones, D.W. Straub Jr., Reconceptualizing system usage: an approach and empirical test, *Inf. Syst. Res.* 7 (3) (2006) 228–246.



- [15] G. Chakraborty, V. Lala, D. Warren, An empirical investigation of antecedents of B2b websites' effectiveness, *J. Interact. Market.* 6 (4) (2002) 51–72.
- [16] W.W. Chin, Issues and opinion on structural equation modeling, *MIS Q.* 2 (1) (1998) vii–xvi.
- [17] N. Chung, C. Koo, The use of social media in travel information search, *Telemat. Inf.* 2 (2) (2015) 215–229.
- [18] N. Chung, H. Lee, S.J. Lee, C. Koo, The influence of tourism website on tourists' behavior to determine destination selection: a case study of creative economy in Korea, *Technol. Forecasting Social Change* 96 (2015) 130–143.
- [19] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, Lawrence Erlbaum Associates, Hillsdale, NJ, 1988.
- [20] L.J. Cronbach, L. Furby, How we should measure change: or should we? *Psychol. Bull.* 74 (1) (1970) 68.
- [21] M.J. Culnan, P.K. Armstrong, Information privacy concerns, procedural fairness, and impersonal trust: an empirical investigation, *Organiz. Sci.* 10 (1) (1999) 104–115.
- [22] D. Ellis, A behavioural approach to information retrieval system design, *J. Doc.* 5 (3) (1989) 171–212.
- [23] eMarketer, 2 Billion Consumes Worldwide to Get Smart(Phnoes) by 2016. 2014.
- [24] C. Fang, J. Lee, M.A. Schilling, Balancing exploration and exploitation through structural design: the isolation of subgroups and organizational learning, *Organiz. Sci.* 1 (3) (2010) 625–642.
- [25] D.R. Fesenmaier, J. Jia, Assessing structure in the pleasure trip planning process, *Tour. Anal.* 5 (1) (2000) 13–27.
- [26] R. Filieri, S. Alguezaui, F. McLeay, Why do travelers trust tripadvisor? Antecedents of trust towards consumer-generated media and its influence on recommendation adoption and word of mouth, *Tou. Manage.* 51 (2015) 174.
- [27] D. Fodness, B. Murray, Tourist information search, *Ann. Tour. Res.* 4 (3) (1997) 503–523.
- [28] C. Fornell, F.L. Bookstein, Two structural equation models: LISREL and PLS applied to consumer exit-voice theory, *Journal of Marketing Research* 9 (4) (1982) 440–452.
- [29] C. Fornell, D.F. Larcker, Evaluating structural equation models with unobservable variables and measurement error, *J. Market. Res.* 18 (1981) 39–50.
- [30] N. Graburn, Tourism: the sacred journey, in: V. Smith (Ed.), *Hosts and Guests: The Anthropology of Tourism*, University of Pennsylvania, Philadelphia, 1989, pp. 21–36.
- [31] U. Gretzel, D.R. Fesenmaier, J.T. O'leary, The transformation of consumer behaviour, in: D. Buhalis, C. Costa (Eds.), *Tourism Business Frontiers: Consumers, Products and Industry*, Elsevier, Burlington, MA, 2006, pp. 9–18.
- [32] U. Gretzel, M. Sigala, X. Xiang, C. Koo, Smart tourism: foundations and developments, *Electron. Markets* 5 (3) (2015) 179–188.
- [33] Z.-L. He, P.-K. Wong, Exploration vs. exploitation: an empirical test of the ambidexterity hypothesis, *Organiz. Sci.* 5 (4) (2004) 481–494.
- [34] C.-I. Ho, Y.-L. Lee, The development of an E-travel service quality scale, *Tour. Manage.* 8 (6) (2007) 1434–1449.
- [35] C.-I. Ho, M.-H. Lin, H.-M. Chen, Web users' behavioural patterns of tourism information search: from online to offline, *Tour. Manage.* 33 (6) (2012) 1468–1482.
- [36] K.F. Hyde, Information processing and touring planning theory, *Ann. Tour. Res.* 5 (3) (2008) 712–731.
- [37] Cheryl B. Jarvis, Scott B. MacKenzie, Philip M. Podsakoff, A critical review of construct indicators and measurement model misspecification in marketing and consumer research, *J. Consum. Res.* 0 (2) (2003) 199–218.
- [38] J. Jeng, D.R. Fesenmaier, Conceptualizing the travel decision-making hierarchy: a review of recent developments, *Tour. Anal.* 7 (1) (2002) 15–32.
- [39] G.R. Jennings, Perceptions on quality tourism experience, in: G.J.A.N.P. Nickerson (Ed.), *Quality Tourism Experiences*, Elsevier Butterworth-Heinemann, Oxford, UK, 2006, pp. 1–22.
- [40] R.A. Johnson, D.W. Wichern, *Applied Multivariate Statistical Analysis*, Pearson, New Jersey, 2007.
- [41] K. Kaplanidou, C. Vogt, A structural analysis of destination travel intentions as a function of web site features, *J. Travel Res.* 5 (2) (2006) 204–216.
- [42] M.J. Kim, C.-K. Lee, M. Bonn, The effect of social capital and altruism on seniors' revisit intention to social network sites for tourism-related purposes, *Tour. Manage.* 53 (2016) 96–107.
- [43] C. Koo, N. Chung, H.-W. Kim, Examining explorative and exploitative uses of smartphones: a user competence perspective, *Inf. Technol. People* 28 (1) (2015) 133–162.
- [44] C.C. Kuhlthau, Developing a Model of the Library Search Process: Cognitive and Affective Aspects (RQ), (1988) , pp. 232–242.
- [45] D. Laureiro-Martínez, S. Brusoni, N. Canessa, M. Zollo, Understanding the exploration–exploitation dilemma: an fmri study of attention control and decision-making performance, *Strateg. Manage. J.* 36 (3) (2015) 319–338.
- [46] K.S. Law, C.-S. Wong, Multidimensional constructs M structural equation analysis: an illustration using the job perception and job satisfaction constructs, *J. Manage.* 5 (2) (1999) 143–160.
- [47] R. Law, S. Qi, D. Buhalis, Progress in tourism management: a review of website evaluation in tourism research, *Tour. Manage.* 1 (3) (2010) 297–313.
- [48] D.A. Levinthal, J.G. March, The myopia of learning, *Strateg. Manage. J.* 14 (1993) 95–112 (Special Issue).
- [49] Y.-H. Li, J.-W. Huang, Exploitative and exploratory learning in transactive memory systems and project performance, *Inf. Manage.* 50 (6) (2013) 304–313.
- [50] H. Liang, N. Saraf, Q. Hu, Y. Xue, Assimilation of enterprise systems: the effect of institutional pressures and the mediating role of top management, *MIS Q.* 1 (1) (2007) 59–87.
- [51] S. Lieber, D. Fesenmaier, Physical and social conditions affecting recreation site preferences, *Environ. Plann. A* 7 (12) (1985) 1613–1625.
- [52] M.K. Lindell, D.J. Whitney, Accounting for common method variance in cross-sectional research designs, *J. Appl. Psychol.* 6 (1) (2001) 114–121.
- [53] F.M. Lord, Elementary models for measuring change, in: C.W. Harris (Ed.), *Problems in Measuring Change*, University of Wisconsin Press, Madison, Wisconsin, 1963.
- [54] X. Luo, Uses and gratifications theory and E-consumer behaviors: a structural equation modeling study, *J. Interact. Advert.* 2 (2) (2002) 34–41.
- [55] C.N. Madu, A.A. Madu, Dimensions of E-quality, *Int. J. Qual. Reliab. Manage.* 19 (3) (2002) 246–258.
- [56] J.G. March, Exploration and exploitation in organizational learning, *Organiz. Sci.* 2 (1) (1991) 71–87.
- [57] L. Moutinho, Consumer behaviour in tourism, *Eur. J. Market.* 1 (10) (1987) 5–44.
- [58] A.M. Munar, J.K.S. Jacobsen, Motivations for sharing tourism experiences through social media, *Tour. Manage.* 43 (2014) 46–54.
- [59] J.D. Neal, M.J. Sirgy, M. Uysal, The role of satisfaction with leisure travel/tourism services and experience in satisfaction with leisure life and overall life, *J. Bus. Res.* 4 (3) (1999) 153–163.
- [60] E. No, J.K. Kim, Comparing the attributes of online tourism information sources, *Comput. Hum. Behav.* 50 (2015) 564–575.
- [61] J.C. Nunnally, *Psychometric Theory*, McGraw-Hill, New York, 1978.
- [62] J.E. Otto, J.B. Ritchie, The service experience in tourism, *Tour. Manage.* 7 (3) (1996) 165–174.
- [63] B. Pan, D.R. Fesenmaier, Online information search: vacation planning process, *Ann. Tour. Res.* 3 (3) (2006) 809–832.
- [64] B. Pan, X. Xiang, R. Law, D.R. Fesenmaier, The dynamics of search engine marketing for tourist destinations, *J. Travel Res.* 0 (4) (2011) 365–377.
- [65] Y.A. Park, U. Gretzel, Success factors for destination marketing web sites: a qualitative meta-analysis, *J. Travel Res.* 6 (1) (2007) 46–63.
- [66] P.A. Pavlou, H. Liang, Y. Xue, Understanding and mitigating uncertainty in online exchange relationships: a principal-agent perspective, *MIS Q.* 1 (1) (2007) 105–136.
- [67] S. Petter, D. Straub, A. Rai, Specifying formative constructs in information systems research, *MIS Q.* 1 (4) (2007) 623–656.
- [68] P.M. Podsakoff, S.B. MacKenzie, J.-Y. Lee, N.P. Podsakoff, Common method biases in behavioral research: a critical review of the literature and recommended remedies, *J. Appl. Psychol.* 8 (5) (2003) 879–903.
- [69] J. Preece, B. Nonnecke, D. Andrews, The top five reasons for lurking: improving community experiences for everyone, *Comput. Hum. Behav.* 0 (2) (2004) 201–223.
- [70] C.M. Ringle, S. Wende, A. Will, Smartpls, SmartPLS, Hamburg, Germany, 2005.
- [71] F.T. Rothaermel, M.T. Alexandre, Ambidexterity in technology sourcing: the moderating role of absorptive capacity, *Organiz. Sci.* 20 (4) (2009) 759–780.
- [72] W. Salisbury, R. Pearson, A. Pearson, D. Miller, Identifying barriers that keep shoppers off the world wide web: developing a scale of perceived web security, *Ind. Manage. Data Syst.* 101 (4) (2001) 165–176.
- [73] S. Schmidt, A.S. Cantallops, C.P. dos Santos, The characteristics of hotel websites and their implications for website effectiveness, *Int. J. Hosp. Manage.* 7 (4) (2008) 504–516.
- [74] M. Sigala, Social networks and customer involvement in new service development (Nsd) the case of www.mystarbucksidea.com, *Int. J. Contemp. Hosp. Manage.* 4 (7) (2012) 966–990.
- [75] C. Smallman, K. Moore, Process studies of tourists' decision-making, *Ann. Tour. Res.* 7 (2) (2010) 397–422.
- [76] H.J. Smith, S.J. Milberg, S.J. Burke, Information privacy: measuring individuals' concerns about organizational practices, *MIS Q.* (1996) 167–196.
- [77] W.W. Smith, X. Li, B. Pan, M. Witte, S.T. Doherty, Tracking destination image across the trip experience with smartphone technology, *Tour. Manage.* 48 (2015) 113.
- [78] M.E. Sobel, Asymptotic confidence intervals for indirect effects in structural equations models, in: S. Leinhardt (Ed.), *Sociological Methodology*, Jossey-Bass, San Francisco, 1982, pp. 290–312.
- [79] Statista, Number of Worldwide Internet Users from 2000 to 2015. 2015.
- [80] M. Subramani, How do suppliers benefit from information technology use in supply chain relationships? *MIS Q.* 8 (1) (2004) 45–73.
- [81] J.E. Swan, L.J. Combs, Product performance and consumer satisfaction: a new concept, *J. Market.* 0 (2) (1976) 25–33.
- [82] R.K. Teas, Expectations, performance evaluation, and consumers' perceptions of quality, *J. Market.* 57 (4) (1993) 18–34.
- [83] R. Tourangeau, L.J. Rips, K. Rasinski, *The Psychology of Survey Response*, Cambridge University Press, Cambridge, England, 2000.
- [84] D.K. Tse, P.C. Wilton, Models of consumer satisfaction formation: an extension, *J. Market. Res.* (1988) 204–212.
- [85] I.P. Tussyadiah, D.R. Fesenmaier, Mediating tourist experiences: access to places via shared videos, *Ann. Tour. Res.* 6 (1) (2009) 24–40.
- [86] J. Urry, *The Tourist Gaze: Leisure and Travel in Contemporary Societies*, Sage Publications, London, 1990.
- [87] G. Van Noort, H.A. Voorveld, E.A. van Reijmersdal, Interactivity in brand web sites: cognitive, affective, and behavioral responses explained by consumers' online flow experience, *J. Interact. Market.* 26 (4) (2012) 223–234.

- [88] D. Wang, S. Park, D.R. Fesenmaier, The role of smartphones in mediating the touristic experience, *J. Travel Res.* 1 (4) (2012) 371–387.
- [89] D. Wang, Z. Xiang, D.R. Fesenmaier, Adapting to the mobile world: a model of smartphone use, *Ann. Tour. Res.* 48 (2014) 11.
- [90] H. Werthner, F. Ricci, E-commerce and tourism, *Commun. ACM* 7 (12) (2004) 101–105.
- [91] Z. Xiang, U. Gretzel, Role of social media in online travel information search, *Tour. Manage.* 1 (2) (2010) 179–188.
- [92] Z. Xiang, V.P. Magnini, D.R. Fesenmaier, Information technology and consumer behavior in travel and tourism: insights from travel planning using the internet, *J. Retail. Consum. Serv.* 22 (2015) 244–249.
- [93] Z. Xiang, D. Wang, J.T. O'Leary, D.R. Fesenmaier, Adapting to the internet: trends in travelers' use of the web for trip planning, *J. Travel Res.* 54 (4) (2015) 0047287514522883.
- [94] C.W. Yoo, Y.J. Kim, G.L. Sanders, The impact of interactivity of electronic word of mouth systems and E-quality on decision support in the context of the E-marketplace, *Inf. Manage.* 52 (4) (2015) 496–505.

**C. Derrick Huang** is an Associate Professor at the Department of Information Technology and Operations Management, College of Business, Florida Atlantic University. Previously, as a practitioner, he held executive-level positions in the area of marketing and strategic planning in a number of high-tech companies. Dr. Huang's research interest lies in the business value and strategic impact of information technology in organizations, and his current focus is on the economics of information security investments, risk management of information systems, and health-care IT. His work has been published in leading journals such as *Decision Sciences Journal*, *Decision Support Systems*, *Information & Management*, *International Journal of Production Economics*, *Communications of the AIS*, *Information Systems Management*, and *Information Systems Frontier*. He holds PhD from the Harvard University.

**Jahyun Goo** is an Associate Professor of MIS at the Florida Atlantic University. His active research areas are IS sourcing, IT and Tourism, interorganizational relation-

ships, health-care IT, and IS Security. His papers have been published in *MIS Quarterly*, *Decision Sciences*, *Information & Management*, *Decision Support Systems*, *Information Systems Journal*, and *Information Systems Frontier*, among others. His work has been recognized from the conferences and publishers as best or outstanding paper awards. He has served for major journals as either a reviewer or a coordinating editor. He holds PhD in MIS from the State University of New York at Buffalo.

**Kichan Nam** is a Visiting Professor at American University of Sharjah in UAE. He received his PhD in Management Information Systems from the State University of New York at Buffalo and was a professor of MIS at Sogang University in South Korea. His major research topics are IT outsourcing, e-business, IT service management, service science, smart tourism, and e-government. His publications are found in major international journals such as *MIS Quarterly*, *Information Systems Research*, *Communications of the ACM*, *Decision Support Systems*, *Journal of Management Information Systems*, etc., and major IS journals in Korea. He was the Chairman of itsMF Korea, Chairman of Korean IT Outsourcing Forum, and Vice President of Korean Society of Management Information Systems. He was a recipient of BK21 and World Class University Project as a project leader in Korea.

**Chul Woo Yoo** is an Assistant Professor in the Department of Information Technology and Operation Management in the College of Business at the Florida Atlantic University. He holds a PhD degree in MIS from the State University of New York at Buffalo. His research interests include information security, human factor in e-business, information privacy, software piracy, and agricultural information system. His works have been published in *Decision Support Systems*, *Information Development*, *Information & Management*, and several proceedings of International Conference on Information Systems and Americas Conference on Information Systems.